Few biological parameters suffice to reproduce measured dynamics: verified parameter subset selection for systems biology models

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Abstract

Biological models contain many unknown rate constants that data rarely suffice to estimate. We present a parameter subset selection method that, starting from published values, finds a small set of biological parameters that reproduces the measured outputs of the full model when only it and the measurement scaling and offset factors are re-estimated. Parameters are ranked by their effect on the outputs, redundant ones are rejected, the reduced model is tested by re-estimation under random perturbations, and unneeded parameters are removed. In 31 of 34 published models with real data a subset, usually of two biological parameters, was sufficient, confirmed by nonlinear re-estimation, and 24 remained sufficient for changes of up to 50% with independent perturbation directions. With the other parameters fixed at their published values, all selected parameters were practically identifiable in 25 models. Random subsets of the same size, or the scaling factors alone, reproduced the full model in 16 and 9 of 28 models. Selected from the same training data, our subsets predicted synthetic data unseen by published values as reliably as Brun's collinearity index and more reliably than pivoted QR and profile likelihood. The mathematical statements underlying these guarantees are verified in the Lean proof assistant.

Keywords: parameter reduction, practical identifiability, sensitivity analysis, ordinary differential equations, systems biology, formal verification

Introduction

Ordinary differential equation (ODE) models are central to quantitative biology, pharmacology and epidemiology, where they describe how interacting molecular species or populations change over time as a function of kinetic parameters θ[1]. Using a model for prediction requires estimating θ from measurements, and this becomes difficult as models grow. The main obstacle is non-identifiability: different parameter values may produce the same measured output, because of model symmetries (structural non-identifiability) or because the data are insufficient (practical non-identifiability)[2–6]. The likelihood then has flat ridges on which optimisers stall and confidence intervals become unreliable[7–9]. Models in public benchmark collections routinely contain 20–300 parameters[10].

The sloppy-model phenomenon offers a structural explanation[11, 12]: the outputs are typically sensitive to a few "stiff" parameter directions and almost insensitive to many "sloppy" ones[13], which suggests that a small subset of parameters may suffice to describe what is actually measured. Global sensitivity indices[14, 15] and profile likelihoods[7, 16, 17] rank parameters by importance or identifiability, and subset-selection methods based on the sensitivity or Fisher information matrix additionally avoid redundant parameters[18]: Fisher-information criteria[19, 20], collinearity indices[21, 22], orthogonalization[23–25], eigenvalue methods[26] and pivoted QR[27, 28], several of which have been compared on signalling and physiological models[26, 29]. All of them return a subset, but, to our knowledge, none tests whether re-estimating the selected subset reproduces the outputs of the full model when all parameters change, and the validity of statements such as "the discarded parameters can be fixed" is rarely made explicit.

This work was motivated by our earlier estimation of a dynamical model of Bandura's social cognitive theory[30–33] with 22 parameters, in which many stable parameter sets fitted the data with widely different quality[34], the typical behaviour when several parameters have nearly identical effects on the measured output.

Here we present a reduction method that works directly on the measured outputs defined in the PEtab format[35], with their experimental conditions, transformations and noise levels. It separates biological parameters from calibration parameters (scalings and offsets of the measurement), which are always re-estimated, and selects only among the biological ones: candidates are ranked by sensitivity, redundant ones are rejected by conditioning tests, the reduced model is accepted only if re-estimating it reproduces the full model under random parameter perturbations, and backward elimination removes every parameter that is not needed. We evaluate the method on 34 benchmark problems with real data[10, 35], on held-out experiments and synthetic data, against four existing criteria and null baselines under a common protocol, and we report its robustness and computational cost. The mathematical statements on which it relies are verified in the Lean 4 proof assistant with Mathlib[36, 37], and we state precisely what is and is not proved.

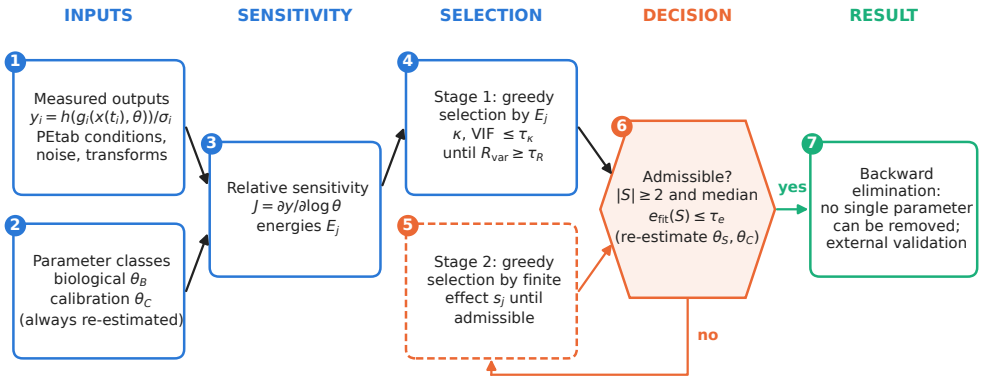


Fig. 1 Overview of the reduction method. Biological parameters are ranked by their effect on the measured outputs and selected with conditioning control (κ and VIF thresholds; stage 1, or stage 2 if needed, dashed). A subset is accepted when re-estimating it reproduces the full model; backward elimination then removes every parameter that is individually dispensable (the result is inclusion-minimal if the criterion is monotone; Theorem 2), and the reduced model is validated on held-out data.

Results

Overview of the method

Figure 1 summarises the procedure; all symbols are defined in Methods (*Measured outputs and parameter classes* to *Admissibility and re-estimation*) and listed in Supplementary Table S2. The input is a PEtab problem: an SBML model, the measured observables with their conditions, times, transformations and noise standard deviations σ_i, and the nominal parameter values θ_0 reported by the original authors. Each measurement defines a scaled output $y_i(\theta) = h(g_i(x(t_i;\theta),\theta))/\sigma_i$, in units of measurement noise. Only biological parameters θ_B (those of the SBML model) are candidates for selection; calibration parameters θ_C are always re-estimated with the selected ones.

The relative sensitivity matrix $J = \partial y/\partial \log \theta$ is computed once at θ_0. In stage 1, biological parameters are added in decreasing order of their sensitivity energy $E_j = \|J_{:,j}\|^2$, a candidate being accepted only if the condition number κ and the largest variance inflation factor (VIF) of the selected columns stay below a threshold, until a fraction τ_R of the energy is captured. A subset S is admissible if, in random scenarios in which all parameters are perturbed, re-estimating θ_S and θ_C with the other parameters at θ_0 gives a median relative fitting error $e_{\text{fit}} \leq \tau_e$, and $|S| \geq 2$. The threshold $\tau_e = \sqrt{1-c^2} = 0.436$ ($c = 0.9$) is the smallest error attainable by rescaling an output change whose cosine similarity with the full change is c (Theorem 1). If stage 1 does not give an admissible subset, stage 2 orders the parameters by their finite-perturbation effect s_j and stops at the first admissible subset. Backward elimination finally removes any parameter whose removal keeps the subset admissible (Theorem 2).

Benchmark: a few biological parameters suffice in most models

We applied the method to 34 PEtab benchmark problems with real measurements[10, 35], covering signalling, metabolism, gene regulation, cell cycle, immunology and epidemiology, with 3 to 294 parameters and 16 to 27,132 measurements (Table 1; Methods, *Models, data and simulation*; Supplementary Note *System-by-system summary*). An admissible subset was found in 31 of the 34 problems (Fig. 2). In 25 of them admissibility was established by nonlinear re-estimation; in the six with the most expensive simulations (Alkan, Bachmann, Chen, Isensee, Lucarelli, Raimundez) it was decided with a first-order approximation of e_{fit} and then checked by nonlinear re-estimation of the final subset, which confirmed it in all six (in Lucarelli with a margin of 4×10^{-5}; Methods, *Admissibility and re-estimation*, *First-order e_{fit} and nonlinear check*). In 28 problems the subset contains two biological parameters, which is also the minimum size enforced by the method ($k_{\min} = 2$): a single biological parameter already reaches the threshold in 15 of 27 tested problems (Methods, *Settings*, *Single-parameter subsets*). The other three subsets contain three, five and ten parameters (Zhao, Bruno, Lucarelli). Counting the calibration parameters, which are re-estimated in every case (up to 79, in Raimundez), the reduced problem has a median of 78% fewer free parameters than the full one (range 33–97%; 88% among biological parameters alone; Methods, *Admissibility and re-estimation*).

All 31 subsets satisfy the conditioning thresholds ($\tau_\kappa = \tau_{\mathrm{VIF}} = 10$; maximum $\kappa = 6.5$, VIF$= 9.8$; Methods, *Sensitivities, energy and conditioning*). In 17 of them the median cosine similarity is also at least c, so that a single rescaling of the reduced response already reaches the threshold, and in two (Alkan, Perelson) the first-order error bound based on the captured energy applies as well; Table 1 gives this level of guarantee for every problem (Methods, *Admissibility and re-estimation*). Stage 1 sufficed in seven problems; in the other 24 the energy ordering selected parameters whose re-estimation could not reproduce the full model, and stage 2 was needed (Methods, *Selection*). The three problems without an admissible subset fail for different reasons: in Crauste, 1% changes of single parameters make the solution grow without bound within the measured horizon, so no local reduction is defined (Methods, *Models, data and simulation*, *Crauste*); in Beer and Lang the search stopped at the conditioning limit with 18 and 13 parameters ($e_{\mathrm{fit}} = 50\%$ and 56%), and in Lang other conditioning thresholds lead to admissible pairs (Methods, *Settings*), so the failure there is one of the greedy search, not of the model (Supplementary Note *System-by-system summary*).

Re-estimation is essential. When the selected parameters are simply set to their perturbed values, the median error e_{rel} exceeds τ_e in 23 of the 31 admissible problems (Fig. 2b; Methods, *Admissibility and re-estimation*); after re-estimation the median e_{fit} is 15% (maximum 43.585%, in Lucarelli, against $\tau_e = 43.589\%$). The selected parameters do not replicate each perturbation; adjusting them compensates for the perturbations of all the other parameters, the property needed when the reduced model is fitted to data.

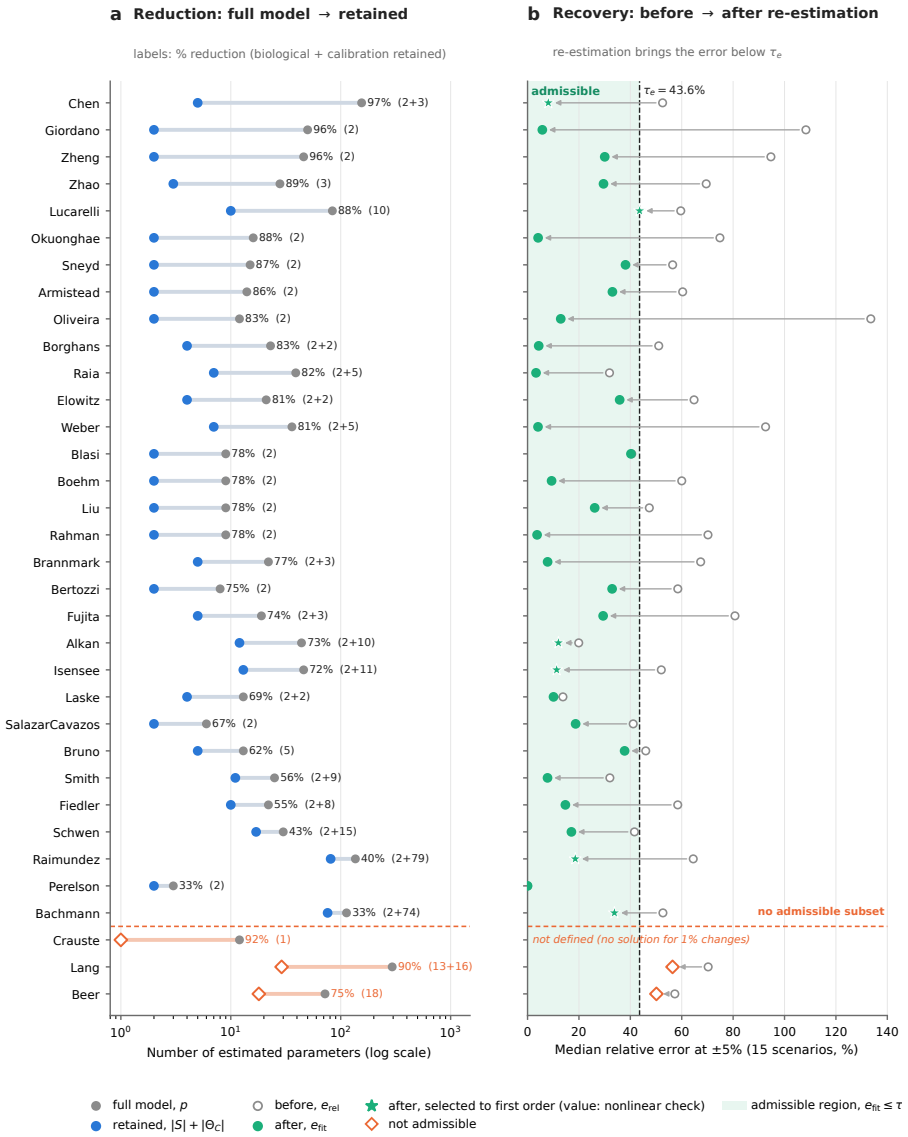


Fig. 2 Parameter reduction and recovery of the full-model output in 34 benchmark problems. Problems are ordered by the reduction achieved; the three without an admissible subset (Crauste, Lang and Beer) are at the bottom. **a**, Estimated parameters of the full model (p, grey) and parameters re-estimated in the reduced model (selected biological plus calibration parameters, blue). **b**, Median relative error over 15 random scenarios at ±5% before (e_{rel}, open circles) and after re-estimating the retained parameters (e_{fit}, green; stars, subsets selected with the first-order e_{fit}, shown with the value of the nonlinear re-estimation of the final subset; for Beer and Lang, first-order value). The shaded region is the admissible region, $e_{\mathrm{fit}} \le \tau_e = 43.6\%$.

Table 1 Per-problem results. p, number of estimated parameters; m, number of measurements; $|\Theta_C|$, calibration parameters (always re-estimated); $|S|$, selected biological parameters; Stage, stage that produced the subset; e_{rel} and e_{fit}, median relative error over 15 scenarios at ±5% without and with re-estimation (admissibility: $e_{\mathrm{fit}} \leq 43.6\%$; italic, not admissible; *, subset selected with the first-order e_{fit}; the value shown is the nonlinear re-estimation of the final subset, except for Beer and Lang, which have no admissible subset (first-order value); †, not defined because the model has no solution on the measured horizon for 1% parameter changes); Pattern, behaviour of the median e_{fit} for perturbation sizes from ±1% to ±50% (A, admissible at all sizes; B, admissible up to at least ±10%, first failing at ±20% or more and not recovering; C, irregular; D, not admissible at ±10% or less; –, not applicable); Level, level of guarantee (1, $e_{\mathrm{fit}} \leq \tau_e$; 2, in addition median $\cos\Delta \geq c$; 3, in addition $R_{\mathrm{var}} \geq \tau_R$); Ident., selected parameters whose profile likelihood in the reduced problem exceeds $\Delta\chi^2 = 3.84$ on both sides within the bounds, out of $|S|$, and in parentheses those for which the hypotheses of the Lean identifiability theorem were satisfied at all evaluated profile points (a numerical check, not a certificate; Table 3); Test χ^2/n, mean squared noise-scaled residual on held-out experiments for the reduced and the full model, both re-estimated on the training experiments (values near 1 mean errors of the size of the noise; –, not evaluable or not run).

System	p	m	$\lvert\Theta_C\rvert$	$\lvert S\rvert$	Stage	e_{rel} (%)	e_{fit} (%)	Level	Pattern	Ident. (Lean)	Test χ^2/n red. / full
Alkan	44	1733	10	2	1	20	12.0*	3	A	2/2 (1)	2.75 / 3.37
Armistead	14	58	0	2	2	60	33.0	1	B	2/2 (2)	1.22 / 1.37
Bachmann	113	541	74	2	2	53	33.8*	1	A	2/2 (0)	4.77 / 7.35
Beer	72	27132	0	18	2	57	*50.1**	–	–	–	–
Bertozzi	8	22	0	2	2	58	32.9	2	B	2/2 (0)	–
Blasi	9	252	0	2	1	41	40.2	2	B	2/2 (2)	–
Boehm	9	48	0	2	2	60	9.4	2	A	2/2 (2)	2.57 / 11.22
Borghans	23	111	2	2	2	51	4.3	1	A	2/2 (0)	0.74 / 0.74
Brannmark	22	43	3	2	2	67	7.8	1	A	2/2 (0)	5.96 / 6.28e+03
Bruno	13	77	0	5	1	46	37.8	2	A	5/5 (5)	0.91 / 208
Chen	155	120	3	2	2	53	8.1*	1	A	2/2 (0)	20.88 / 17.71
Crauste	12	21	0	1	2	–	*n.d.*†	–	–	–	–
Elowitz	21	58	2	2	1	65	35.8	1	B	2/2 (0)	1.50 / 3.89
Fiedler	22	72	8	2	2	58	14.7	1	A	2/2 (0)	11.21 / 9.32
Fujita	19	144	3	2	2	81	29.4	1	A	0/2 (0)	5.94 / 8.03
Giordano	50	313	0	2	2	108	5.8	2	A	0/2 (0)	<0.01 / <0.01
Isensee	46	687	11	2	1	52	11.3*	1	A	2/2 (0)	1.08 / 1.08
Lang	294	9600	16	13	2	70	*56.4**	–	–	–	–
Laske	13	42	2	2	1	14	10.1	2	A	2/2 (1)	0.76 / 1.87
Liu	9	100	0	2	2	47	26.1	2	A	2/2 (0)	0.95 / 1.02
Lucarelli	84	1755	0	10	2	60	43.585*	1	C	10/10 (4)	1.66 / 3.34
Okuonghae	16	92	0	2	2	75	4.1	2	A	1/2 (0)	4.67e4 / 1.22e5
Oliveira	12	120	0	2	2	134	12.9	2	A	1/2 (0)	912 / 8.13e4
Perelson	3	16	0	2	1	0	0.0	3	A	0/2 (0)	<0.01 / <0.01
Rahman	9	23	0	2	2	70	3.7	2	A	2/2 (2)	<0.01 / 0.03
Raia	39	205	5	2	2	32	3.3	2	A	2/2 (0)	8.22 / 5.10
Raimundez	136	627	79	2	2	64	18.6*	1	A	2/2 (0)	1.17 / 1.43
SalazarCavazos	6	18	0	2	2	41	18.7	2	A	2/2 (0)	102 / 124
Schwen	30	286	15	2	2	42	17.1	2	A	2/2 (0)	8.20 / 8.40
Smith	25	62	9	2	2	32	7.8	2	A	1/2 (0)	1.50e4 / 1.50e4
Sneyd	15	135	0	2	2	56	38.1	1	A	2/2 (2)	2.22 / 2.56
Weber	36	135	5	2	2	93	4.1	1	A	2/2 (0)	1.00 / 15.95
Zhao	28	82	0	3	2	70	29.6	2	A	3/3 (1)	0.35 / 0.35
Zheng	46	60	0	2	2	95	30.0	1	A	2/2 (2)	1.58 / 20.82

The selected parameters are also identifiable

A subset that reproduces the full model need not be determined by the data; a selected rate constant could, for example, compensate for a measurement scaling factor. We therefore computed the profile likelihood of every selected parameter in the reduced problem, with $S \cup \Theta_C$ free and the other parameters at θ_0 (Methods, *Identifiability of the selected parameters*). In 25 of the 31 problems all selected parameters were practically identifiable, that is, their 95% profile-likelihood interval was bounded on both sides within the range of the study (65 of the 74 selected parameters; Table 1, column Ident.). The exceptions were Fujita, where a selected rate constant is confounded with a scaling factor; Giordano and Perelson, whose recorded noise levels are far larger than the residuals; Okuonghae and Oliveira, in which the profile of one parameter could not be simulated; and Smith, whose optimum for one parameter lies at the bound. In the profile likelihoods of the full model, with all parameters free (computed for the comparison of criteria; Methods, *Comparison with other selection criteria*, *Profile likelihood*), only 20 of the 52 selected parameters that could be evaluated were identifiable: the data do not determine all parameters of these models but, once the others are fixed at their published values, they determine the few that the method selects. In every problem whose profiles could be evaluated, identifiability was predicted by the profiled information $I_j = 1/[(A^\top A)^{-1}]_{jj}$, the quantity on which a sufficient condition verified in Lean is based (Methods, *Identifiability of the selected parameters*, *Formal condition*). Because only the selected and calibration parameters are free, the check took a median of 34 s per problem.

Prediction of held-out experiments

We split the measurements of each problem into training and test sets, by experimental condition (21 problems) or, with a single condition, by time (10 problems; Methods, *External validation*), fitted the reduced model (θ_S and θ_C) and the full model (all parameters) to the training data from θ_0, and evaluated both on the test data. With the non-selected parameters at θ_0, the reduced model predicted the held-out experiments at least as accurately as the full model (test χ^2/n within 5% or lower) in 26 of the 29 comparable problems (Fig. 3a); the exceptions were Raia, Fiedler and Chen. Where the full model overfitted the training data the difference was large (for example Brannmark, test χ^2/n 6.0 against 6,277; Bruno, 0.9 against 208).

This comparison favours the reduced model, because θ_0 was estimated from all data, test experiments included, and the reduced model inherits that information through the parameters it keeps fixed. Two checks quantify this (Fig. 3c; Methods, *External validation*, *Leakage-free nested validation*): θ_0 alone predicted the test data better than the reduced model, by more than 5%, in 20 of 29 problems (median ratio 0.90), and a nested reduced model, with the non-selected parameters fixed at a full-model fit to the training data instead of θ_0, predicted worse than the θ_0-anchored one in 21 of 29 (median ratio 1.35). The advantage of the reduced model on real data therefore reflects the value of anchoring at published estimates, not a property of the selected subset alone.

To measure prediction without this confound, we generated synthetic data from a known truth θ^* within $\pm 20\%$ of θ_0 in every estimable parameter, with noise of the recorded size, and repeated the experiment with 20 seeds (Methods, *External validation, Synthetic experiment*; Supplementary Note *Synthetic prediction experiment*); θ_0 has not seen these data, although it still indicates the region of the truth, so the test is local. Among the cases in which the parameter change exceeded the noise, the reduced model reached a relative prediction error $e_{\mathrm{pred}} \leq \tau_e$ in 56% (178 of 320) and predicted at least as well as the full model in 57% (160 of 282). The outcome was stable within each problem (Fig. 3b): the reduced model predicted better in almost every seed of the problems in which the full model overfits (Weber 20 of 20 seeds, Zheng 18 of 20, Isensee 12 of 12) and worse in almost every seed of others (Oliveira 0 of 20, Fujita 1 of 18). On data that θ_0 has not seen, we found no overall predictive advantage of the reduced models over the full models, and equivalence was not tested; the reduced models predicted better where the full model overfitted the training data. Success decreased with the distance of the truth from θ_0 (64%, 54% and 51% at $\pm 5\%$, $\pm 20\%$ and $\pm 50\%$, ten seeds) and grew with the size of the true change in noise units (Methods, *External validation, True signal*).

Comparison with existing selection criteria

We compared our subsets with those of four existing criteria: pivoted QR of the sensitivity matrix in the standard subset-selection form of Golub, Klema and Stewart[27] (the original variant, pivoted QR of the normalised matrix itself, is reported in the Supplementary Information); the collinearity index of Brun *et al.*[21]; Morris screening over $[\theta_0/10,\ 10\theta_0]$[15]; and profile likelihood[7, 16], retaining the parameters with the narrowest confidence intervals. Every criterion chose the same number $k = |S|$ of biological parameters from the training outputs, and every subset was evaluated with the same three tests: reproduction of the full model (the admissibility test, at $\pm 5\%$ and at $\pm 20\%$), prediction of the held-out experiments and prediction of synthetic data (Methods, *Comparison with other selection criteria*; Table 2, Fig. 4). Two null baselines re-estimated only the calibration parameters ($k = 0$) or k biological parameters drawn at random (20 draws). Our main subsets were selected on all measured outputs at θ_0; to make the comparison equitable, we repeated our selection on the training outputs alone and passed these subsets through the same tests (Methods, *Comparison with other selection criteria, Selection on the training outputs and other families of truths*).

On real data the criteria were practically indistinguishable (Fig. 4c; Methods, *Statistics*): Wilcoxon tests over problems found no significant difference between our subsets and those of Brun, Morris, profile likelihood, pivoted QR, random subsets or our own subsets selected on the training outputs ($P = 0.16$–0.46), and the calibration-only baseline was slightly better (median ratio of test χ^2/n 0.99, $P = 0.003$, $n = 28$). As expected for sloppy models[11], many small subsets fit the training data and predict similarly when the other parameters stay at θ_0, so this test does not discriminate between criteria.

The criteria differed in whether their subsets reproduced the full model (Fig. 4a). At $\pm 5\%$, our subsets were admissible in all 28 comparable problems, against 26 for

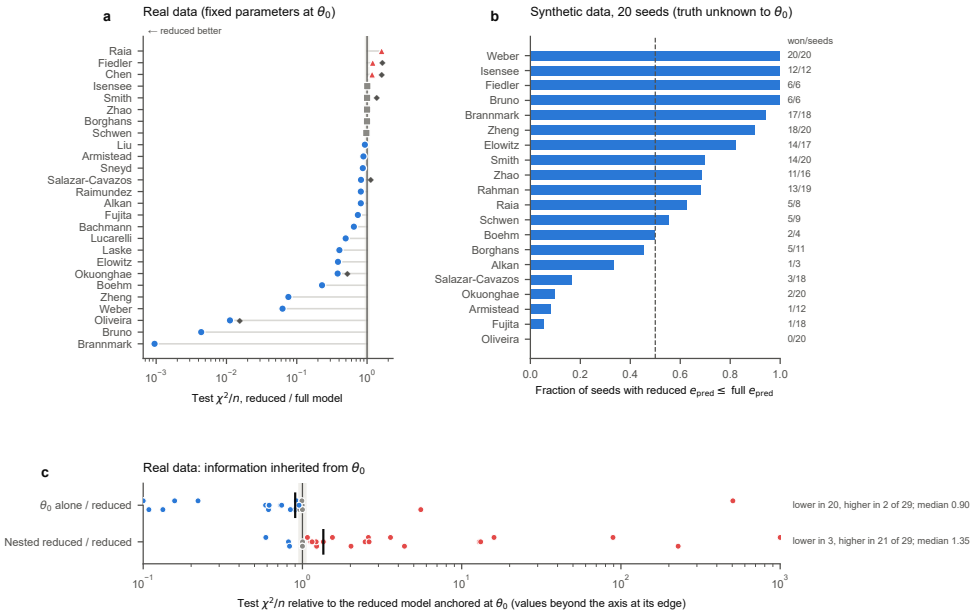


Fig. 3 Prediction of held-out experiments by the reduced and the full model. a, Real data: ratio of the test χ^2/n (mean squared noise-scaled residual on the held-out experiments) of the reduced model to that of the full model re-estimated on the same training data, on a logarithmic scale ($n = 26$ problems), with the non-selected parameters kept at θ_0. Left of the vertical line (circles) the reduced model predicts better; the grey band marks equivalence within 5% (squares); triangles mark worse predictions; ♦ flags a reduced test χ^2/n above 10, where a favourable ratio does not imply a good absolute fit. **b**, Synthetic data from a known truth, which θ_0 has not seen: for each problem, fraction of the 20 seeds in which the reduced model predicted at least as well as the full model (smaller e_{pred}), among the seeds in which the parameter change exceeded the noise (seeds won / seeds with signal on the right; problems with fewer than three such seeds are not shown). Problems are ordered by this fraction. **c**, Real data, information inherited from θ_0: test χ^2/n of the published estimate θ_0 without re-estimation, and of the nested reduced model (non-selected parameters fixed at a full-model fit to the training data instead of θ_0), each relative to the reduced model anchored at θ_0; one point per problem ($n = 29$), vertical bar at the median; blue, lower by more than 5%; red, higher by more than 5%; grey band, within 5%. Problems without a training/test split (Bertozzi, Blasi), without a full re-estimation (in **b**, Chen and Raimundez) or with noise-free data (**a**: Giordano, Perelson, Rahman) are not shown. Per-problem values and scatter plots are given in Supplementary Notes *Prediction of held-out experiments* and *Leakage-free nested validation.*

Brun, 23 for Morris, 18 of 27 for profile likelihood and 17 for pivoted QR (exact McNemar test, Methods, *Statistics*; Holm-adjusted $P = 0.5$, 0.13, 0.012 and 0.004). The null baselines show that the test is demanding: re-estimating the calibration parameters alone was admissible in 9 of 28 problems, and random subsets of the same size in a median of 16 of 28 (14–19 over the 20 draws). This comparison favours our method, which selects by this very test; selected on the training outputs alone, our subsets were admissible in 24 of 27 problems, close to Brun's. At ±20%, a level used by none of the criteria, the counts were 27, 25, 23, 18 of 27 and 16 of 28 ($P = 0.5$, 0.25, 0.023 and 0.004), and 23 of 27 for our training-output subsets; the same directions were rescaled, so this tests the size of the change, not new directions.

In prediction on synthetic data (Fig. 4b), over 281 cases with signal in 24 problems, our subsets reached $e_{\text{pred}} \leq \tau_e$ in 56% of the cases, Brun's in 55%, Morris's in 48%, those of profile likelihood in 39% and those of pivoted QR in 30%; resampling whole problems (Methods, *Statistics*), the advantage was 8 percentage points over Morris (95% confidence interval 2–16; Holm-adjusted $P = 0.010$), 17 over profile likelihood (7–28; $P = 0.002$) and 26 over pivoted QR (12–40; $P < 0.001$), against 1 over Brun (−1 to +3; $P = 0.43$). Because these truths belong to the family of perturbations that our method uses to decide admissibility, we repeated the comparison with log-uniform truths within a factor of two of θ_0 and with sparse truths in which 15% of the parameters change by ±50% (Methods, *Comparison with other selection criteria*, *Selection on the training outputs and other families of truths*). The advantage over profile likelihood and pivoted QR persisted (log-uniform: 50% against 37% and 24%, $P = 0.018$ and < 0.001; sparse: 54% against 40% and 39%, $P = 0.008$ and 0.019), that over Morris held for sparse truths (46%, $P = 0.003$) but not for log-uniform ones (48%), and no family separated our method from Brun's. Selected on the training outputs, our subsets reached the threshold in 51%, 48% and 55% of the cases of the three families, again more often than those of profile likelihood and pivoted QR ($P \leq 0.046$) and not differently from Brun's. The median prediction error did not differ significantly between criteria (Supplementary Note *Comparison of selection criteria on synthetic data*). What distinguishes our method is therefore not its ranking, which it shares with Brun's collinearity index and with orthogonalization methods[23, 25], but the verification that follows it: the subset is tested by re-estimation on the outputs of interest, failures are reported, and, chosen from the same data, our subsets predicted unseen data as reliably as the best existing criterion and more reliably than pivoted QR and profile likelihood, a difference that we observed and that the verification does not guarantee.

Robustness to the size of parameter changes

Admissibility is decided with perturbations of ±5%. Keeping each final subset fixed, we repeated the evaluation at ±1, 5, 10, 20, 30, 40 and 50% (15 scenarios per level) and classified the median e_{fit} into four patterns (A, admissible at all levels; B, failing first at ±20% or more; C, irregular; D, failing at ±10% or less; Methods, *Robustness patterns*). When the same fifteen directions are rescaled at every level, 26 of the 31 problems follow pattern A (Fig. 5), four pattern B (Armistead, Bertozzi, Blasi, Elowitz, whose errors are already 33–40% at ±5%) and one pattern C (Lucarelli, on

Table 2 Comparison of selection criteria and null baselines. All criteria choose the same number $k = |S|$ of biological parameters, from the training outputs except for our main subsets (selected on all outputs), and are evaluated with the same protocol and the fixed scenario set. Admissible: problems in which re-estimating the subset reproduces the full model (median $e_{\mathrm{fit}} \leq \tau_e$) for parameter changes of ±5% and ±20%; for random subsets, median over the 20 draws; for calibration only, without the rule $|S| \geq 2$. Synthetic: fraction of the cases with signal in which $e_{\mathrm{pred}} \leq \tau_e$, for truths uniform within ±20% of θ_0 (281 cases common to all criteria, 24 problems, 20 seeds; difference from our method with a 95% confidence interval from resampling problems), log-uniform within a factor of two (217 cases, 25 problems) and sparse (147 cases, 24 problems), ten seeds each. Real data: median ratio of the test χ^2/n to that of our method (below 1, better than ours) and Wilcoxon P over problems (unadjusted; exploratory). Pivoted QR is the standard Golub–Klema–Stewart subset selection; the earlier variant is reported in the Supplementary Information. Per-problem values are given in the Supplementary Notes on the comparison of selection criteria.

	Admissible		Synthetic: $e_{\mathrm{pred}} \leq \tau_e$				Real data
Criterion	±5%	±20%	Uniform	Ours − other (95% CI)	Log-unif.	Sparse	χ^2 ratio (P)
Ours (all outputs)	**28/28**	**27/28**	56%	–	50%	54%	–
Ours (training)	24/27	23/27	51%	+5 [0, 12]	48%	55%	1.00 (0.17)
Brun	26/28	25/28	55%	+1 [−1, 3]	50%	52%	1.00 (0.27)
Morris	23/28	23/28	48%	+8 [2, 16]	47%	46%	1.00 (0.46)
Profile likelihood	18/27	18/27	39%	+17 [7, 28]	37%	40%	1.00 (0.44)
Pivoted QR	17/28	16/28	30%	+26 [12, 40]	24%	39%	1.00 (0.16)
Random (20 draws)	16/28	16/28	–	–	–	–	1.00 (0.27)
Calibration ($k = 0$)	9/28	9/28	–	–	–	–	0.99 (0.003)

the threshold at all levels); none follows pattern D. With independent directions at each level, the more conservative design, 24 problems follow pattern A, one pattern B and six pattern C; the nine problems that change are those whose median error is close to the threshold at large perturbations (Supplementary Note *Robustness to the size of parameter changes*). The median hides individual failures: at ±5% the median pass rate is 93%, and 13 problems reproduce the full model in every valid scenario. Most reductions thus remain valid well beyond the perturbation used for selection; those with a small margin at ±5% fail for large changes.

Sensitivity to method settings and computational cost

We re-ran the complete method on the 26 small and medium problems changing one setting at a time (Methods, *Settings*). The 25 admissible subsets were unchanged with finite-difference steps of 0.1% or 5%, with either alternative set of integrator tolerances and with τ_R between 0.80 and 0.95; the perturbation size (±1% or ±10%) changed one subset, and conditioning thresholds of 5 or 30 changed three or four subsets without changing the number of admissible problems. The error threshold has the largest influence: with $\tau_e = 0.10$, 0.20, 0.30 and 0.50, 10, 15, 18 and 23 subsets were unchanged, and $\tau_e = 0.10$ requires a mean of 3.7 biological parameters against 2.1–2.5 for τ_e between 0.30 and 0.50. The threshold is therefore an explicit accuracy–parsimony trade-off, not a tolerance validated for any biological

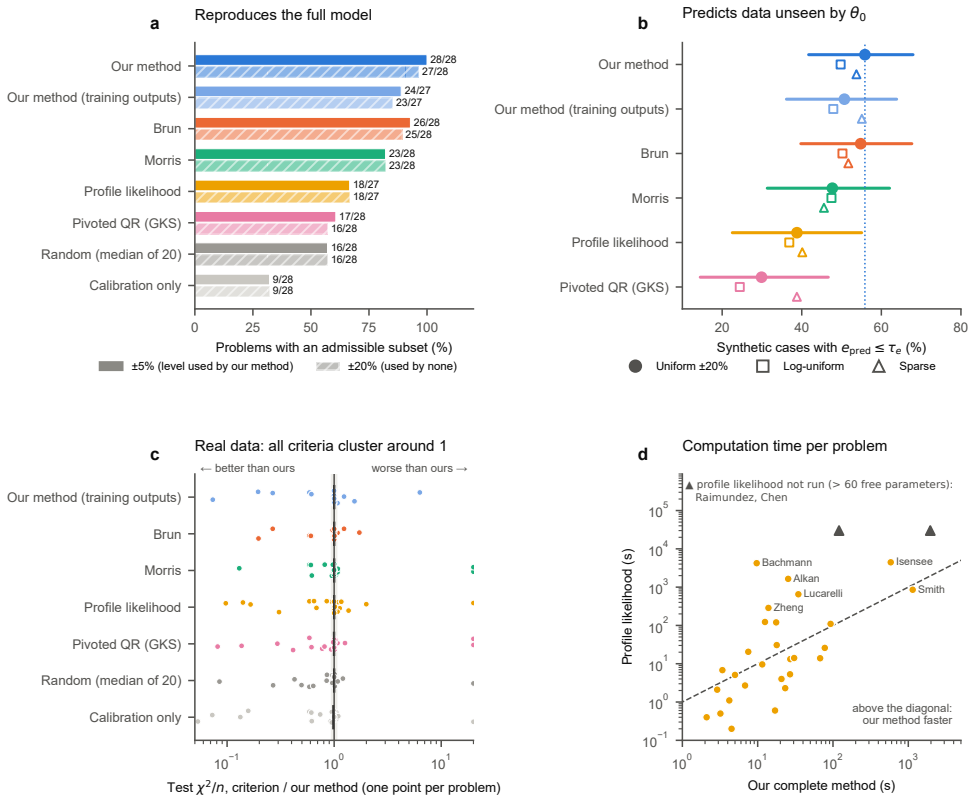


Fig. 4 Our method compared with four existing selection criteria and null baselines. a, Number of problems in which the subset of each criterion is admissible, i.e. its re-estimation reproduces the full model, for parameter changes of ±5% (the level used by our method) and ±20% (used by none); 28 problems (27 for profile likelihood and for our subsets selected on the training outputs); random subsets, median over 20 draws. **b**, Fraction of synthetic cases with signal in which the subset predicts the truth within τ_e, for three families of truths (filled circles, uniform within ±20% of θ_0, with 95% confidence intervals from resampling problems; squares, log-uniform; triangles, sparse); dotted line, our method with uniform truths. **c**, Real data: ratio of the test χ^2/n of each criterion to that of our method, one point per problem, vertical bar at the median (log scale; grey band, ±5%; values outside the axis are drawn at its edge); all criteria cluster around 1. **d**, Computation time of our complete method and of profile likelihood per problem (log scales); points above the diagonal are faster with our method; problems in which profile likelihood was not run (Chen and Raimundez) are shown at the top.

application: e_{fit} is relative to the output change caused by the perturbation, and in units of the recorded noise its median remaining error per measurement was below 0.1σ in 17 of 31 problems and below σ in 29 (Supplementary Table S33). The random seed and the number of scenarios were not varied.

The number of sensitivity simulations grows linearly with the number of parameters ($3p$), they are independent and can run in parallel, and the admissibility test adds 15 re-estimations per candidate subset, whose cost depends on the simulation time and on the number of retained parameters rather than on p. Run

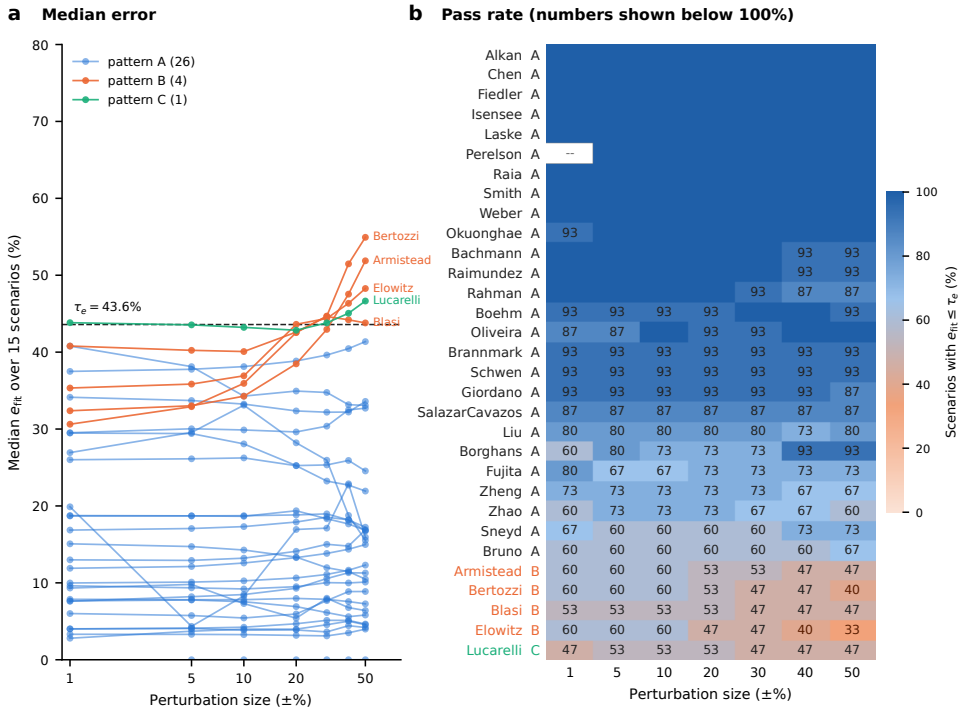


Fig. 5 Robustness of the selected subsets to the size of parameter changes. a, Median relative error after re-estimation (e_{fit}) over 15 random scenarios per level as a function of the perturbation size, for the final subset of each of the 31 admissible problems, coloured by robustness pattern (A, admissible at all levels; B, first failing at ±20% or more; C, irregular); dashed line, $\tau_e = 43.6\%$. **b**, Pass rate: percentage of the scenarios with $e_{\mathrm{fit}} \leq \tau_e$, with the value written when it is below 100%. The same fifteen random directions are used at every level. –, no valid scenario (Perelson at ±1%, where the output change is below the numerical noise floor).

from scratch on a shared server with the sensitivities in parallel, the selection took from a few seconds to 33 min per problem (median 19 s; about 3 min for Lang, with 294 parameters). Profile likelihood was as fast or faster on small problems, because the admissibility test has a fixed cost (our method was faster than its first round in 12 of 28 problems), but much slower on large ones: 70 min on 60 cores for Bachmann (90 free parameters; 40 h of processor time) against 10 s, 5.2 h for Isensee (110 h of processor time) against 10 min, and an estimated 10^3–10^5 h of processor time for Raimundez and Chen, which we did not run. The method itself cost about 23 h of processor time for the 34 problems: 1.3 h for the selection and 21.3 h for the nonlinear verification of the final subset in the six problems selected to first order, almost all of it in Chen (12.8 h), Raimundez (4.1 h) and Isensee (3.9 h). The remaining ∼611 h were spent on the validations, robustness tests and comparisons carried out for this paper; all analyses reported here together took about 634 h of processor time, summed over cores; because they ran in parallel on servers with

88–256 cores, the elapsed time was much shorter (Methods, *Computational cost*; Supplementary Note *Computing time of the analyses*).

Formal verification of the mathematical foundations

The statements on which the method relies are verified in Lean 4 with Mathlib in 32 modules with 309 theorems and lemmas, together with 28 benchmark models; all proofs compile with a single command, without unproved assumptions (`sorry`) or additional axioms (Methods, *Formal verification*; Code availability; Supplementary Tables S4 and S7). Table 3 states, for each step of the method, what is proved, when the statement applies to a numerical result and to which results of this study it applies. Two statements are central:

Theorem 1. *For vectors $d \neq 0$ and $b \neq 0$ in a real inner-product space,*

$$\min_{\lambda \in \mathbb{R}} \frac{\|d - \lambda b\|}{\|d\|} = \sqrt{1 - \cos^2(d, b)}.$$

Theorem 2. *Let $\mathcal{A}$ be any admissibility predicate on finite sets and S an admissible set. Backward elimination (removing one element at a time whenever the remaining set is still admissible, until no element can be removed) returns an admissible subset of S from which no element can be removed without losing admissibility. If, in addition, $\mathcal{A}$ is monotone (every superset of an admissible set is admissible), no proper subset of the result is admissible.*

Theorem 1 links τ_e to the cosine: whenever $\cos \Delta \geq c$, rescaling the reduced response gives a relative error of at most $\sqrt{1 - c^2}$; the converse does not hold, and admissibility is decided by the computed e_{fit}. The first part of Theorem 2 makes no assumption on the predicate and so holds for it as implemented; this is the sense in which the selected subsets are minimal. Minimality is relative to the criterion, which includes $|S| \geq 2$, and is not minimal cardinality. The second part requires monotonicity, which we checked empirically on all 660 nested pairs of subsets evaluated during the selections (no violation; Methods, *Selection, Monotonicity log*). Further verified statements cover the conditioning tests, re-estimation, the transfer of error bounds to transformed outputs and the first-order approximation (Methods, *Formal verification, Further verified statements*). A sufficient condition for practical identifiability (theorem `practically_identifiable_grad`) bounds the profile-likelihood interval of a parameter through the profiled information I_j, the residual gradient at the optimum and a bound ε on the nonlinear remainder over the admissible range. Its linear version (Methods, *Identifiability of the selected parameters, Formal condition*) agreed with the profile classification for all 72 selected parameters whose profile could be evaluated. With the nonlinear remainder estimated at the computed profile points, the hypotheses of the theorem were satisfied at every evaluated point for 24 parameters, and never for a parameter whose profile is not identifiable; this is a numerical check, not a certificate. Over the admissible set, as the theorem requires, the remainder bound holds only near the optimum: with two

variants proved in Lean that need the bound only on the tail of the profile, sampling and adversarial search (Methods, *Identifiability of the selected parameters, Hypotheses over the admissible set*) found the hypotheses satisfied for 9 of the 74 parameters within a factor of 1.5 of the optimum, 7 within a factor of 2 and 1 over the whole range of the study. Identifiability over the range of the study therefore rests on the computed profiles; with twice as many profile points, the same 65 parameters were identifiable. The verification is local and concerns implications: the quantities to which the theorems are applied are computed in floating point, and a conclusion of Table 3 applies to a result only when the model hypotheses are verified in Lean and the computed quantities satisfy the stated inequalities (Supplementary Table S5).

Discussion

For most published systems biology models with real data, the measured outputs can be reproduced by re-estimating only two biological parameters together with the measurement calibration parameters, and, with the other parameters fixed at their published values, the data also determine those two parameters. In the terminology of system identification this is a parameter subset selection[27] whose result is checked twice: the subset suffices to reproduce the outputs of the full model and its parameters are practically identifiable. With all parameters free, most selected parameters are not identifiable, which is the sloppy-model picture[11, 12]; our statement is more concrete because it refers to the outputs actually measured, under the actual conditions and noise levels, and it is checked by re-estimation and profiles rather than inferred from the sensitivity spectrum. It is a statement about output sufficiency and identifiability, not about predictive superiority: on data that the published estimate has not seen, the reduced models predicted neither consistently better nor consistently worse than the full models, and their advantage on real data reflects the information contained in the published values.

The comparison with existing criteria clarifies what the method adds. Subsets chosen by pivoted QR, Brun's collinearity index, Morris screening, profile likelihood or at random predict held-out experiments about equally well when the other parameters stay at their published values. The criteria differ in whether their subsets actually reproduce the full model: random subsets and calibration-only re-estimation failed in 43% and 68% of the problems, profile likelihood and pivoted QR in 33% and 39%, and the subsets chosen by sensitivity with a redundancy test (Brun's and ours) rarely failed. Chosen from the same training data, our subsets predicted unseen data as reliably as Brun's and more reliably than those of pivoted QR and profile likelihood, for three families of synthetic truths. The ranking itself is not new—E_j is the diagonal of the Fisher information matrix of the scaled outputs and, up to a constant, the squared importance measure of Brun *et al.*[21], and orthogonalization methods[23–25] combine the same magnitude with a greedy redundancy test, much like stage 1—and energy alone sufficed in only seven problems, because it measures how strongly a parameter moves the outputs, not whether adjusting it can compensate for changes in other parameters. What the method adds is the verification by re-estimation on the outputs of interest, the reduction by backward

elimination until no parameter can be removed, and the fact that failures become visible (Beer, Lang, Crauste) instead of returning an untested subset.

The method is practical: its sensitivity simulations grow linearly with the number of parameters and run in parallel, its selection took from seconds to about half an hour per problem on models with up to 294 parameters, and the nonlinear verification of the final subset at most 13 h of processor time in the problem with the slowest simulations, and its subsets were unchanged by the finite-difference step, the integrator tolerances and τ_R, whereas the error threshold τ_e sets an explicit accuracy–parsimony trade-off. The mathematical statements behind each step are proved in Lean 4; a statement applies to a particular result when the computed quantities satisfy the inequalities listed in Table 3. Other approaches answer different questions: likelihood-based observability analysis[9] assesses prediction uncertainty after calibration, Bayesian approaches[38] require many more simulations, structural identifiability analysis[2–4] addresses ideal noise-free data, manifold-based methods[39] rewrite the model in terms of emergent parameter combinations, balanced truncation[40, 41] reduces state variables, and hierarchical optimisation[42] eliminates calibration parameters without selecting among biological ones.

The study has limitations. (i) The analysis is local: sensitivities are computed at θ_0, assumed to be a good fit, admissibility is tested around it, and the synthetic truths are generated around it. (ii) Because the non-selected parameters stay at θ_0, predictions on real held-out data benefit from information that θ_0 already contains. (iii) The minimum subset size of two is a design choice; one parameter already reaches the threshold in about half of the problems. (iv) The error threshold $c = 0.9$ is a design choice relative to the perturbation, and in Giordano, Perelson and Lucarelli a $\pm 5\%$ perturbation changes the outputs by less than the measurement noise, so their relative admissibility has little consequence for the data. (v) The explicit finite-perturbation certificates of the formal development (Methods) are too conservative to certify any scenario at $\pm 5\%$ in our models (constants 10^2–10^{24}), so admissibility rests on the local theorems and on the numerical test. (vi) The method assumes smooth models, Gaussian noise with the recorded standard deviations and re-estimable calibration parameters; in five problems the recorded noise does not match the scatter of the data, one problem with 9,169 conditions was too expensive to analyse on its measured outputs, and three problems had no admissible subset.

In practice the method can be used as a preprocessing step: before calibrating a large model, select a small set of biological parameters, verify that re-estimating them reproduces the full model and that their profiles are bounded, keep the rest at literature values, and check the reduced model on held-out experiments. Natural extensions are forward or adjoint sensitivities for genome-scale models[43], sensitivities at several operating points and the use of the selected subsets in experimental design[44].

Methods

Models, data and simulation

We used the problems of the PEtab benchmark collection[10, 35] that provide real measurements, an SBML model and nominal parameter values (34 problems; Table 1). Each problem specifies the observables g_i, their transformation h (linear, log or $\log_{10}$), the noise standard deviation σ_i of each measurement, the experimental conditions (parameter or initial-value overrides) and, where applicable, pre-equilibration conditions. Models were simulated with libRoadRunner[45] (CVODE, relative and absolute tolerances 10^{-7} and 10^{-10}). Initial assignments that depend on parameters were re-evaluated for every parameter vector, condition-specific overrides were applied before simulation, pre-equilibration was run to steady state before the main simulation, and models with discontinuous inputs were integrated piecewise between discontinuities. Simulated observables were compared with the reference simulations distributed with the benchmark wherever available (Supplementary Note *Verification of the simulator*, Supplementary Table S10). *Crauste.* To establish why no local reduction exists for this problem, each of its 12 estimated parameters was changed by $\pm 1\%$ one at a time and the states were simulated over the measured horizon of 28 days (`crauste_blowup.py`); the solution grows without bound before the end of the horizon in 9 of these 24 changes and in 7 of the 15 admissibility scenarios at $\pm 5\%$, with the same counts for relative integrator tolerances from 10^{-10} to 10^{-5}. One further benchmark problem with real data (Froehlich, 4,231 parameters and 9,169 experimental conditions) was analysed only on its state trajectories, because each evaluation of its measured outputs requires about 9,000 simulations; it is not counted in the 34 problems (see Computational cost).

Measured outputs and parameter classes

For a parameter vector $\theta \in \mathbb{R}^p$ (in the units of the benchmark, excluding noise parameters), the vector of scaled outputs is $y(\theta) \in \mathbb{R}^m$ with entries

$$y_i(\theta) = \frac{h\big(g_i(x(t_i;\theta,c_i),\theta)\big)}{\sigma_i}, \qquad i = 1,\dots,m, \tag{1}$$

where $x(t;\theta,c)$ is the state trajectory under condition c and (t_i, c_i) are the time and condition of measurement i. Parameters that appear in the SBML model are biological (Θ_B); parameters that appear only in the observation functions and not in the model or the condition table (scalings and offsets) are calibration parameters (Θ_C). Only biological parameters are candidates for selection; calibration parameters are always re-estimated.

Sensitivities, energy and conditioning

The relative sensitivity matrix $J \in \mathbb{R}^{m \times p}$ is computed by central differences,

$$J_{:,j} = \frac{y(\theta_0 + \delta\theta_{0,j} e_j) - y(\theta_0 - \delta\theta_{0,j} e_j)}{2\delta}, \tag{2}$$

which approximates $\partial y / \partial \log \theta_j$. The energy of parameter j is $E_j = \|J_{:,j}\|^2 = (J^\top J)_{jj}$ and the fraction of energy captured by a subset $S \subseteq \Theta_B$ is $R_{\mathrm{var}}(S) = \sum_{j \in S} E_j / \sum_{j \in \Theta_B} E_j$. For a subset S, the columns of J_S are centred and scaled to unit variance, giving a matrix F whose Gram matrix is proportional to the correlation matrix R_S of the sensitivity columns; the condition number is $\kappa(S) = \sigma_{\max}(F)/\sigma_{\min}(F)$ and the largest variance inflation factor is $\mathrm{VIF}(S) = \max_j [R_S^{-1}]_{jj}$[18, 21]. Scaling removes differences of scale between parameters, so that κ and VIF measure only the angles between sensitivity directions. Because R_S is a principal submatrix of the correlation matrix of all columns, both quantities can only increase when parameters are added. The statements of the formal development on the linearised least-squares problem are formulated for columns scaled to unit length without centring, $Z = J_S \,\mathrm{diag}(\|J_{:,j}\|)^{-1}$; computed in that form for the final subsets (`conditioning_check.py`), $\kappa(Z) \leq 6.6$ and $\mathrm{VIF}(Z) \leq 10$ in 30 of the 31 subsets, whereas in Rahman $\kappa(Z) = 12.4$ and $\mathrm{VIF}(Z) = 39$, so the formal statement on the linearised problem does not apply to that subset.

Selection

Greedy selection. Candidates are visited in a given order. A candidate j is accepted if $\kappa(S \cup \{j\}) \leq \tau_\kappa$ and $\mathrm{VIF}(S \cup \{j\}) \leq \tau_{\mathrm{VIF}}$; otherwise it is discarded permanently. Because κ and VIF can only increase when columns are added, a discarded candidate would also be rejected later. *Stage 1* visits biological parameters in decreasing order of E_j and stops when $R_{\mathrm{var}}(S) \geq \tau_R$. *Stage 2*, used if the stage-1 subset is not admissible, visits biological parameters in decreasing order of the finite-perturbation effect $s_j = \|y(\theta_0 + \eta\theta_{0,j} e_j) - y(\theta_0)\|$ and stops at the first admissible subset. *Backward elimination* visits the parameters of the selected subset and removes each one whose removal leaves the subset admissible (between one and five parameters were removed in five problems; Supplementary Table S1). *Monotonicity log.* All subsets evaluated during a selection and their admissibility are stored; for every nested pair $S \subset T$ of evaluated subsets we count the violations of monotonicity (S admissible, T not), which bear on the second part of Theorem 2.

Admissibility and re-estimation

Admissibility is evaluated on N random scenarios $\theta^{(k)} = \theta_0 \odot (1 + \varepsilon u^{(k)})$, with $u^{(k)}$ uniform in $[-1, 1]^p$. For each scenario let $\Delta y_{\mathrm{full}} = y(\theta^{(k)}) - y(\theta_0)$. The relative error without re-estimation is $e_{\mathrm{rel}} = \|\Delta y_{\mathrm{full}} - \Delta y_S\| / \|\Delta y_{\mathrm{full}}\|$, where $\Delta y_S = y(\theta^{(k)}_{S \cup \Theta_C}, \theta_{0,\overline{S \cup \Theta_C}}) - y(\theta_0)$ is the output change when only the parameters in $S \cup \Theta_C$ take their scenario values and all others stay at θ_0, and the cosine similarity

is $\cos\Delta = \langle \Delta y_{\text{full}}, \Delta y_S \rangle / (\|\Delta y_{\text{full}}\| \|\Delta y_S\|)$. The relative error after re-estimation is

$$e_{\text{fit}} = \min_{\theta_S, \theta_C} \frac{\|y(\theta_S, \theta_C, \theta_{0,\overline{S}}) - y(\theta^{(k)})\|}{\|\Delta y_{\text{full}}\|}, \tag{3}$$

where $\theta_{0,\overline{S}}$ denotes the nominal values of all other biological parameters. The minimisation is carried out in $\log|\theta|$ within $[|\theta_0|/10,\ 10|\theta_0|]$ by a trust-region reflective least-squares method (SciPy[46], at most $100(|S| + |\Theta_C| + 1)$ function evaluations), from two starting points (the scenario value and θ_0); the error without re-estimation is also included, so that $e_{\text{fit}} \le e_{\text{rel}}$. Scenarios whose full-model output change is below $\max(10^{-10}\|y(\theta_0)\|, 10^{-6})$, or in which the simulation of the full model fails, are discarded, and the number of valid scenarios is reported; this set does not depend on S. A scenario in which the simulation with only $S \cup \Theta_C$ at their scenario values fails is kept and counts as not reproduced ($e_{\text{rel}} = \infty$). (The selection originally discarded such scenarios from the median of e_{rel}; re-running all selections with the fixed set changed no subset in any of the 34 problems.) Parameters with a negative nominal value keep their sign in the scenarios and in the re-estimation, which acts on $\log|\theta|$; a parameter with nominal value zero (four in Lang and two in Raimundez) is represented by 10^{-12}, so its relative perturbations and its column of J are negligible and, like any parameter with a zero or negligible column, it is in practice never selected (Supplementary Note *Random scenarios and large problems*). Supplementary Table S3 reconciles the parameter counts used throughout (estimated, biological, calibration and noise parameters, and the free parameters of each analysis). S is admissible if

$$|S| \ge k_{\min} \quad \text{and} \quad \bar{e}_{\text{fit}}(S) = \underset{k}{\text{median}}\, e_{\text{fit}}^{(k)}(S) \le \tau_e = \sqrt{1 - c^2}, \tag{4}$$

the median being taken over the valid scenarios. The reduction in the number of free parameters reported in Results is $1 - (|S| + |\Theta_C|)/p$, and $1 - |S|/|\Theta_B|$ among biological parameters. Because e_{rel} bounds the exact e_{fit} from above, the implementation also accepts a subset whose median e_{rel} is at most τ_e, which makes re-estimation unnecessary; with nonlinear re-estimation $e_{\text{fit}} \le e_{\text{rel}}$ in every scenario, so the rule $\min(\text{median}\, e_{\text{fit}}, \text{median}\, e_{\text{rel}}) \le \tau_e$ reduces to the median of e_{fit}, and it differs only for the first-order e_{fit} described below. For each admissible subset we also report a level of guarantee: level 1 when only this condition holds, level 2 when in addition the median $\cos\Delta$ is at least c, and level 3 when in addition $R_{\text{var}}(S) \ge \tau_R$; these levels are reported, not required. *First-order e_{fit} and nonlinear check.* For the eight problems in which the cost of simulation made repeated re-estimation impractical (marked in Table 1), the first-order version $e_{\text{fit,lin}} = \min\{e_{\text{rel}}, \min_c \|\Delta y_{\text{full}} - J_{S\cup\Theta_C} c\| / \|\Delta y_{\text{full}}\|\}$, evaluated scenario by scenario, was used during the selection. This equals the minimum relative error attainable if the model response were exactly linear in parameter changes; for a truly linear model it equals the exact e_{fit}, while for nonlinear models it is an approximation that replaces per-scenario optimisation with a single projection; its absolute error is bounded by a second-order term (Supplementary Table S7), but no bound on the

relative error was evaluated for the individual scenarios. The admissibility decision during the selection is therefore made by linear algebra alone. For the six of these problems with an admissible subset, the final subset was afterwards re-estimated nonlinearly in the same 15 scenarios, with the same starting points, bounds and evaluation budget as for the other problems, the scenarios being distributed over processes (`efit_nonlinear.py`); it confirmed admissibility in all six (median e_{fit}, the values reported in Table 1 and Fig. 2: Alkan 0.120, Bachmann 0.338, Chen 0.081, Isensee 0.113, Lucarelli 0.43585 against $\tau_e = 0.43589$, Raimundez 0.186; Supplementary Note *Nonlinear check of the first-order problems*). The columns of J for the calibration parameters are computed together with those of the biological parameters, because the first-order re-estimation uses $J_{S\cup\Theta_C}$.

Settings

Unless stated otherwise: $\delta = 0.01$; $\eta = 0.1$; $\tau_R = 0.89$; $\tau_\kappa = \tau_{\mathrm{VIF}} = 10$; $N = 15$ scenarios; $\varepsilon = 0.05$; random seed 42; $c = 0.9$, so $\tau_e = 0.436$; $k_{\min} = 2$. The sensitivity analysis changed one setting at a time: $\delta \in \{0.1\%, 5\%\}$; relative/absolute integrator tolerances $10^{-9}/10^{-12}$ and $10^{-6}/10^{-8}$; $\varepsilon \in \{0.01, 0.10\}$; $\tau_R \in \{0.80, 0.85, 0.95\}$; $\tau_\kappa = \tau_{\mathrm{VIF}} \in \{5, 30\}$; and $\tau_e \in \{0.10, 0.20, 0.30, 0.50\}$ (Supplementary Note *Sensitivity to the method settings*). It was run on the 26 problems that could be re-run completely fifteen times in reasonable time; the other eight (Alkan, Bachmann, Beer, Chen, Isensee, Lang, Lucarelli and Raimundez) are those with the most expensive simulations. To relate e_{fit} to the precision of the data, the remaining error of each scenario was also expressed per measurement in units of the recorded noise, $e_{\mathrm{fit}}\|\Delta y_{\mathrm{full}}\|/\sqrt{m}$ (the outputs are already divided by σ_i), and its median over the scenarios was reported. *Single-parameter subsets.* Single-parameter subsets were tested on 27 admissible problems (all except the four with the most expensive simulations, Bachmann, Chen, Isensee and Raimundez): the ten biological parameters with the largest E_j and the ten with the largest s_j were each tested alone, with Θ_C re-estimated, on the same 15 scenarios at $\pm 5\%$ and without the requirement $|S| \geq 2$; the admissible candidate with the smallest median e_{fit} was then validated like the selected subset; in the synthetic experiment the selected subset predicted better than the best single parameter in 9 of these 15 problems, whereas on real data the single parameter had the lower test χ^2/n in 10 of 15 (Supplementary Note *Single-parameter subsets*). In the scan of settings, the perturbation size changed the subset only in Zhao, and admissibility only for Perelson at $\pm 1\%$, where the output change falls below the numerical noise floor. The admissibility test uses the median over $N = 15$ scenarios rather than requiring all of them to pass, because individual scenarios near the boundary of the parameter domain can fail through ODE nonlinearity; six problems pass with 8–9 of the 15 scenarios. For Lucarelli the median e_{fit} lies within 10^{-3} of τ_e to first order and within 4×10^{-5} after nonlinear re-estimation, so its admissibility is borderline, although stable across perturbation sizes (43–47% from $\pm 1\%$ to $\pm 50\%$). To relate the size of a perturbation to the noise, we computed, for each admissible problem, the root mean square per measurement of Δy_{full} at $\pm 5\%$ (in units of σ); it is below 1 in Giordano, Perelson and Lucarelli (Supplementary Table S33).

External validation

Measurements were split into training and test sets by experimental condition (every third condition after sorting, or the second of two conditions, used for testing) or, for problems with a single condition, by time (the last third of the distinct measurement times). Parameters that only affect test conditions were excluded from re-estimation. In two problems a split was not possible: in Bertozzi the selected parameters act only in the test conditions, and Blasi has a single condition and a single time point. All real-data numbers come from a single run of each fit; the re-estimation of over-parameterised full models can end in different local optima[47]. In five problems even θ_0 has a test $\chi^2/n > 10$ (Chen, Okuonghae, Oliveira, Salazar-Cavazos, Smith); our simulation agrees with the reference simulation of the benchmark for Chen, Salazar-Cavazos and Smith, whereas no reference is available for Okuonghae and Oliveira. A large χ^2/n can reflect recorded noise levels that understate the scatter of the data, a misspecified model or inadequate parameter values, and our analysis does not distinguish between these causes. The reduced model (re-estimating $S \cup \Theta_C$) and the full model (re-estimating all parameters) were fitted to the training data from θ_0 with the same least-squares method and bounds, $\hat{\theta} = \arg\min \sum_{i \in \text{train}} (y_i(\theta) - \tilde{y}_i)^2$ over the free parameters of each model within $[|\theta_0|/10, 10|\theta_0|]$, and evaluated by the mean squared noise-scaled residual $\chi^2/n = \frac{1}{n} \sum_{i \in \text{test}} (y_i(\hat{\theta}) - \tilde{y}_i)^2$, where $\tilde{y}_i$ are the scaled measurements. *Synthetic experiment.* In the synthetic experiment, the truth θ^* had each estimable parameter drawn uniformly within $\pm 20\%$ of θ_0 (parameters acting only in test conditions stayed at θ_0), data were simulated with Gaussian noise of standard deviation σ_i, and $e_{\text{pred}} = \|y_{\text{test}}(\hat{\theta}) - y_{\text{test}}(\theta^*)\| / \|y_{\text{test}}(\theta^*) - y_{\text{test}}(\theta_0)\|$; the effect of the parameter change was considered larger than the noise when χ^2/n of θ_0 on the synthetic test data was at least 2. The experiment was repeated with 20 seeds (42–61), each defining a different truth and noise realisation; the full model was not re-estimated for the two problems with more than 100 free parameters after removing those that act only in the test conditions (Chen and Raimundez; Bachmann has 101 parameters that affect the outputs but 84 free ones, and its full model was re-estimated). The experiment was repeated with truths within $\pm 5\%$ and $\pm 50\%$ of θ_0 (seeds 42–51, ten seeds), and compared with the same ten seeds at $\pm 20\%$. With these amplitudes, the reduced model reached $e_{\text{pred}} \le \tau_e$ in 64% and 51% of the cases with signal at $\pm 5\%$ and $\pm 50\%$ (54% at $\pm 20\%$ on the same seeds) and predicted at least as well as the full model in 67% and 47%. *True signal.* The true signal of a case is the root mean square, over the test measurements, of $y_{\text{test}}(\theta^*) - y_{\text{test}}(\theta_0)$, which is in units of σ because the outputs are scaled by the noise; it is computed from the truth, without noise, whereas the criterion "larger than the noise" above uses the noisy synthetic data. At $\pm 20\%$, the reduced model reached $e_{\text{pred}} \le \tau_e$ in 1%, 24%, 44% and 75% of the cases whose true signal was below 0.3σ, between 0.3σ and σ, between σ and 3σ, and above 3σ; across seeds, the success rate ranged from 31% to 76% and the fraction of cases predicted at least as well as by the full model from 40% to 73%. Problems whose test χ^2/n at θ_0 was below 0.01 (Giordano, Perelson, Rahman) reproduce the data to far below the recorded noise level and are not shown in Fig. 3a. Test errors were compared through the ratio of the reduced to the reference χ^2/n; ratios between 0.95 and 1.05 were considered equivalent. In the synthetic

experiment, a case is a pair (problem, seed); the reduced model predicted at least as well as the full model when $e_{\mathrm{pred}}^{\mathrm{red}} \le e_{\mathrm{pred}}^{\mathrm{full}}$, and the success rate is the number of cases with signal and $e_{\mathrm{pred}} \le \tau_e$ divided by the number of cases with signal. *Leakage-free nested validation.* For the leakage-free nested validation (Supplementary Note *Leakage-free nested validation*), the full model was first re-estimated on the training data (yielding $\hat{\theta}_{\mathrm{train}}$), and the reduced model was then re-estimated with the non-selected parameters fixed at $\hat{\theta}_{\mathrm{train}}$ and search bounds derived from $\hat{\theta}_{\mathrm{train}}$, for 29 problems (Bertozzi and Blasi excluded; for Chen the columns of the finite-difference Jacobian were evaluated in parallel and the integrator tolerances were $10^{-6}/10^{-8}$).

Comparison with other selection criteria

Each criterion selected the same number $k = |S|$ of biological parameters as our method, from the training data, and the calibration parameters of the problem were always added. *Pivoted QR*: the standard subset selection of Golub, Klema and Stewart[28], as used for parameter selection by Vélez-Reyes and Verghese[27]: the sensitivity matrix of the training outputs at θ_0, with columns normalised to unit length, is decomposed as $U\Sigma V^{\top}$, and the first k pivots of the column-pivoted QR decomposition of $V_k^{\top}$ (the k dominant right singular vectors) are selected. A simpler variant, the first k pivots of the column-pivoted QR decomposition of the normalised matrix itself, in which the first pivot is decided by rounding and column order when several columns have nearly equal norm, was used in an earlier version of this analysis and is reported in Supplementary Note *Pivoted QR selection*. *Brun*[21]: dimensionless sensitivities $s_{ij} = \theta_{0,j}\,\partial y_i/\partial\theta_j$ on the n training outputs (forward differences, relative step 10^{-4}), importance $\delta_j^{\mathrm{msqr}} = \|s_{:,j}\|/\sqrt{n}$, collinearity index $\gamma_K = 1/\sqrt{\lambda_{\min}(\tilde{s}_K^{\top}\tilde{s}_K)}$, where $\tilde{s}_K$ is s_K with its columns scaled to unit length, and determinant measure $\rho_K = \det(s_K^{\top}s_K)^{1/(2k)}$; among all k-subsets K of the 15 parameters with the largest δ_j^{msqr}, the one with $\gamma_K \le 10$ and the largest ρ_K is selected (the smallest γ_K if none satisfies the bound). *Morris*[15]: each biological parameter is represented by $x_j \in [0,1]$, mapped linearly onto $\log|\theta_j| \in [\log|\theta_{0,j}| - \log 10,\ \log|\theta_{0,j}| + \log 10]$; ten trajectories (seed 42) on the grid $\{0, \frac{1}{3}, \frac{2}{3}, 1\}$ change one parameter at a time, in random order, by $\Delta = \frac{2}{3}$, giving elementary effects $\mathrm{EE}_j = \|y(x+\Delta e_j) - y(x)\|/\Delta$ on the training outputs, and the k parameters with the largest mean effect μ_j^* over the trajectories are selected. *Profile likelihood*[7, 16]: the full model was first re-estimated on the training data; each biological parameter was then fixed at five values on each side of the optimum (up to a factor of 10, within the same bounds as all other fits) and all other parameters were re-estimated; whenever a point of a profile improved the optimum by more than one unit of χ^2, the full model was re-estimated from that point and the profiles were repeated (at most three times; ten for Raia). A parameter was classified as identifiable when its profile exceeded $\Delta\chi^2 = 3.84$ (95%, one degree of freedom) on both sides, and the k parameters with the narrowest confidence intervals were selected, the width being $w_j = t_j^- + t_j^+$, where $t_j^{\pm}$ is the displacement in $\log|\theta_j|$ at which $\Delta\chi^2$ first exceeds 3.84 on each side, interpolated linearly between profile points (∞ if it is not exceeded); ties are broken in favour of the larger of the smaller maxima of $\Delta\chi^2$ on the two sides. It was run for the 26 admissible problems with at

most 60 free parameters and, in a separate run with the profile points distributed over 60 cores, for Bachmann and Lucarelli (90 and 72 free parameters); it was not run for Chen and Raimundez because of its cost. Extrapolating from Bachmann with the simulation time and the numbers of free parameters and profile points, Raimundez (126 free parameters, about 570 profile points, 0.3 s per simulation) would need of the order of 10^3 h of computing time and Chen (155 free parameters, about 1,520 points, 35 s per simulation) of the order of 10^5 h (Supplementary Note *Profile likelihood*). The criteria were implemented as they are usually applied and therefore differ in some numerical settings (data used to select, derivative scheme and step, candidate set, normalisation and computational budget); these differences are listed in Supplementary Note *Brun and Morris selections*. The *calibration-only* baseline re-estimated Θ_C alone ($k = 0$), and the *random* baseline used 20 random k-subsets of the biological parameters (seeds 0–19); for the latter, the number of admissible problems was counted for each draw and its median over the draws is reported. Every subset was then evaluated with the external validation described above (for random subsets, the median test χ^2/n over the draws), with the admissibility test of our method (the same 15 scenarios and the same rules, with the fixed scenario set described in Admissibility and re-estimation, at $\varepsilon = 0.05$ and, with the same directions $u^{(k)}$, at $\varepsilon = 0.20$; for $k = 0$ the rule $|S| \geq 2$ was not applied) and with the synthetic experiment (20 seeds). *Selection on the training outputs and other families of truths.* Our main subsets were selected on all measured outputs at θ_0, whereas the other criteria selected on the training outputs; to remove this difference, our selection was repeated on the rows of the training set only (`REPRO_OUTPUTS=train`), and the resulting subsets were evaluated with the same three tests. The synthetic truths of the main experiment, $\theta^* = \theta_0 \odot (1 + 0.2u)$, belong to the same family of perturbations that our method uses to decide admissibility; the comparison of criteria was therefore repeated, with ten seeds (42–51), for two other families: log-uniform truths, $\theta^*_j = \theta_{0,j}\, 10^{v_j}$ with v_j uniform in $[-0.3, 0.3]$, and sparse truths, in which a random 15% of the estimable parameters change by $\pm 50\%$ and the others stay at θ_0. The problems entering each comparison follow from these requirements. Admissibility was compared in 28 of the 31 admissible problems: Bertozzi was excluded (see Statistics), Blasi has no training/test split on which the other criteria could select, and Chen was not included because the selections of pivoted QR, Brun and Morris on its training outputs require 1,734 simulations of 35 s each (about 17 h of computing time) before any re-estimation. Profile likelihood was compared in 27 of them (all except Raimundez). The synthetic comparison used the 24 problems with at least one seed in which all five criteria had a result and the parameter change exceeded the noise: of the 28, Raimundez lacks the profile-likelihood subset, and Giordano, Laske and Lucarelli have no seed with signal. The computation time of the complete method was measured by running it from scratch, without cached results, with the sensitivity computations distributed over 16–64 cores; all timings were obtained on a 128-core server shared with other computations and are therefore upper bounds.

Statistics

Comparisons between criteria were made over problems. For real data and for the synthetic prediction error, the statistic of each problem was the logarithm of the ratio of the errors of two criteria (for synthetic data, its median over seeds) and the two-sided Wilcoxon signed-rank test was applied. For the fraction of synthetic cases reaching $e_{\mathrm{pred}} \le \tau_e$, the success rate of a criterion is $\hat{\pi} = \sum_s a_s / \sum_s n_s$, where n_s is the number of cases with signal of problem s that are common to all criteria and a_s the number of them with $e_{\mathrm{pred}} \le \tau_e$; 95% confidence intervals of the difference $\hat{\pi}_{\mathrm{ours}} - \hat{\pi}_m$ are the 2.5 and 97.5 percentiles of the difference recomputed in 10,000 resamples of whole problems with replacement (seed 0), so that seeds of the same problem are not treated as independent. The two-sided p value of the bootstrap is twice the smaller of the fractions of resampled differences at or below and at or above zero; in the Wilcoxon test, problems with equal errors were discarded (zero method of Wilcoxon). Admissibility, one outcome per problem, was compared with the exact McNemar test: if b problems are admissible only with our subset and c only with the other, $P = \min\{1,\, 2\sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 2^{-(b+c)}\}$. The comparison of admissibility and of the synthetic success rate with the four existing criteria is the primary analysis; within each family of four comparisons (admissibility at one perturbation level, or synthetic success rate) p values were adjusted with the Holm procedure, $\tilde{p}_{(r)} = \max_{q \le r} \min\{1, (5-q)\, p_{(q)}\}$ for the ordered values $p_{(1)} \le \cdots \le p_{(4)}$, and the comparisons on real data are exploratory. The absence of a significant difference is not evidence of equivalence; no equivalence margin was tested. Bertozzi was excluded from the comparison of admissibility because its selected parameters act only in the test conditions; selected on the training outputs, our method finds no admissible subset of two parameters there (a single parameter, `RO_CA`, has $e_{\mathrm{fit}} = 0.08$). Exact P values are reported; for the bootstrap the resolution is 2×10^{-4}.

Identifiability of the selected parameters

For each problem with an admissible subset, the reduced problem ($S \cup \Theta_C$ free, other parameters at θ_0) was fitted to all measurements from θ_0, with the method and bounds of the external validation. For each selected parameter θ_j, the profile

$$\Delta\chi^2_j(t) = \min_{\log|\theta_j| = \log|\hat{\theta}_j| + t} \chi^2(\theta) - \chi^2(\hat{\theta}), \qquad \chi^2(\theta) = \sum_i \big(y_i(\theta) - \tilde{y}_i\big)^2, \tag{5}$$

where $\hat{\theta}$ is the optimum and the minimum is taken over the other free parameters of the reduced problem, was computed at ten displacements t per side, with $|t|$ from 0.05 to 4.6; points outside the bounds $[|\theta_0|/10, 10|\theta_0|]$ of θ_j were not used, and if a point improved the optimum the reduced problem was re-fitted from it and the profile repeated (at most three rounds). θ_j was classified as identifiable if $\Delta\chi^2$ exceeded 3.84 on both sides, practically non-identifiable if on one side, and flat otherwise[7]; a robust variant scaled the threshold by $\max(1, \chi^2_{\mathrm{min}}/\mathrm{d.f.})$. *Formal condition.* Let $r(\delta)$ be the vector of noise-scaled residuals of all measurements when the logarithms of the free parameters move by δ from the optimum, $r_0 = r(0)$, so that

$\chi^2 = \|r\|^2$, and let A be its sensitivity matrix at the optimum (central differences, step 0.01 in $\log|\theta|$). The profiled information I_j was computed as the squared norm of the component of the column of A for θ_j orthogonal to the other columns, which equals $1/[(A^\top A)^{-1}]_{jj}$ when $A^\top A$ is invertible; it is the output change, in noise units, that a unit change of $\log|\theta_j|$ produces when the other free parameters compensate as well as possible. The residual gradient γ is the norm of the projection of r_0 onto the range of A (zero at an exact optimum), and ε is the largest nonlinear remainder $\|r(\delta) - r_0 - A\delta\|$ over the computed profile points. The Lean theorem `practically_identifiable_grad` states that if $|\langle r_0, A\delta\rangle| \le \gamma\|A\delta\|$, the nonlinear remainder is at most ε, and $t^* = (\varepsilon + \gamma + \sqrt{\gamma^2 + 3.84 + 2\|r_0\|\varepsilon + 2\gamma\varepsilon})/\sqrt{I_j}$ is smaller than the distances d_j^- and d_j^+ from the optimum to the bounds of $\log|\theta_j|$, then θ_j is practically identifiable. With $\gamma = \varepsilon = 0$, t^* reduces to the Wald half-width $\sqrt{3.84/I_j}$, and the linear version of the condition, used in Results, is $\sqrt{3.84/I_j} < \min(d_j^-, d_j^+)$; the remainder can be evaluated only at points, so this condition was checked at the computed profile points. *Hypotheses over the admissible set.* Because the theorem requires the remainder bound over the whole admissible range, not only at profile points, we proved two variants (`practically_identifiable_grad_tail` and `practically_identifiable_rel_tail`) that need the bound only on the tail of the profile beyond the guaranteed half-width, either in absolute form ($\le \varepsilon$) or relative to the linear change ($\le \eta\|A\delta\|$ with $\eta < 1$). For every selected parameter and side, the tail bounds were estimated over the region of the free parameters within a factor $r \in \{10, 5, 3, 2, 1.5\}$ of the optimum (and within the bounds of the study) from 512 Sobol points, the profile points in the region and an adversarial search (L-BFGS-B maximisation of the remainder, or of the ratio, on the tail, started from the worst sampled points and repeated until the half-width was stable); a parameter satisfies the hypotheses in the region when one of the two conditions holds on both sides. As a check of the profile resolution, all profiles were recomputed with 20 instead of 10 displacements per side, and the profiles of the subsets selected on the training outputs were computed with the same procedure. The complete mathematical derivation, the profiles of every problem and the values of all quantities for every parameter are given in Supplementary Note *Identifiability of the selected parameters*. The profiles were computed for all 31 problems with an admissible subset: those of Bachmann, Chen, Isensee and Smith on a 256-core server, that of Raimundez on a 128-core server, and the other 26 on an 8-core workstation (libRoadRunner 2.10.0 on the servers and 2.9.2 on the workstation).

Robustness patterns

For each admissible problem, the final subset was kept fixed and e_{fit} was evaluated at perturbation sizes $\varepsilon \in \{0.01, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50\}$ with 15 scenarios per level. The fifteen directions $u^{(k)} \in [-1, 1]^p$, with independent uniform components, are drawn once per problem with a fixed seed (42) and reused at every level, $\theta^{(k)} = \theta_0 \odot (1 + \varepsilon u^{(k)})$; the directions are stored with the results. To check that the patterns do not depend on these particular directions, the evaluation was repeated with new directions at each size (seed 42 at ±5%, which reproduces the selection, and

$42 + 1000i$ at the i-th other level). Let $\bar{e}(\varepsilon)$ be the median e_{fit} at level ε and ε_f the smallest level with $\bar{e}(\varepsilon_f) > \tau_e$. The pattern is A if no such level exists; C (irregular) if $\bar{e}(\varepsilon) \le \tau_e$ again at a level above ε_f; otherwise B if $\varepsilon_f \ge 0.20$ and D if $\varepsilon_f \le 0.10$; the pass rate is the fraction of valid scenarios with $e_{\mathrm{fit}} \le \tau_e$. As in the selection, e_{fit} was computed to first order for the problems marked in Table 1. For Chen, between 8 and 13 of the 15 scenarios per level could be simulated in the run shown in Fig. 5 (7 to 13 over all our runs, because some simulations fail in a run-dependent way); the remaining scenarios failed to integrate and were discarded. A scenario is valid when it could be simulated and its output change exceeds the numerical noise floor; all fifteen scenarios were valid in 28 of the 31 problems, and only 8, 4 and 2 in Chen, Isensee and Perelson. Among the 26 problems of pattern A, 9 pass every scenario at every level, whereas 7 (Borghans, Bruno, Fujita, Liu, Sneyd, Zhao, Zheng) fall to a pass rate of 60–73% at some level. The full distribution of the error for every problem and level is given in Supplementary Tables S11 and S12 and Supplementary Fig. S4.

Computational cost

The cost of the selection is dominated by the computation of J ($2p$ simulations) and of the stage-2 ordering (p simulations); the selection, the conditioning tests and the first-order e_{fit} only use linear algebra on J (Supplementary Note *Computational cost*). The computation time of the complete method was measured by running it from scratch, without cached results, with the sensitivity computations distributed over 16–64 cores of a 128-core server shared with other computations, so the times are upper bounds. Times are given as elapsed (wall-clock) time when the computation of a problem ran in parallel, and as processor time, the sum over tasks, when it consisted of independent single-core tasks (scenarios, seeds, profile points). For Armistead, Lang and Raimundez the timing run preceded a correction of the scenario generation that changed their subsets; the sensitivity simulations, which dominate the cost, are not affected. Chen, with 35 s per simulation, was the slowest problem (33 min). In the six problems selected with the first-order e_{fit}, the nonlinear re-estimation of the final subset in the 15 scenarios took 4 min (Lucarelli) to 12.8 h (Chen) of processor time (Alkan 8 min, Bachmann 23 min, Isensee 3.9 h, Raimundez 4.1 h). Profile likelihood was much slower on problems with expensive simulations (Alkan 27 min against 25 s for our method; Zheng 5 min against 14 s; Isensee 5.2 h, or 110 h of processor time, against 10 min). For Bachmann (90 free parameters in the profile, 84 in the synthetic experiment after removing those that act only in test conditions; Supplementary Table S3) and Lucarelli (72), profile likelihood took 70 and 11 min with the profile points distributed over 60 cores (40 and 9 h of computing time), against 10 and 35 s for our method; Chen and Raimundez were processed by our method in 33 and 2 min. Selecting a subset took less than ten minutes in 32 of the 34 problems. All analyses reported in this paper together took about 634 h of processor time, the sum over all cores of the times recorded by each task in the final runs (earlier runs superseded by corrections of the code, repeated runs and the compilation of the Lean proofs are not included); because the analyses ran in parallel on servers with 88–256 cores and on an 8-core workstation, the elapsed time was much shorter. 87% of the processor time was spent in Isensee, Chen, Raimundez, Smith and Bachmann; the

largest items were profile likelihood of the full model (193 h), the identifiability profiles (131 h, of which 102 h for the check with 20 points per side), the tests of the subsets of all criteria (86 h) and the synthetic experiments (130 h over all amplitudes, families and criteria). The check of the identifiability hypotheses over the admissible set took a further 3 h of elapsed time on an 88-core server (Supplementary Note *Computing time of the analyses*). On state trajectories, the 4,231-parameter Froehlich model required 12,693 simulations and 2.6 h for the sensitivities; for models of this size on measured outputs, forward sensitivity equations[48] compute J in a single integration of an augmented system, whereas adjoint methods are efficient for gradients of a scalar objective but not for the full output Jacobian needed here.

Formal verification

The formal development uses Lean 4 (version 4.23.0) and Mathlib[36, 37] and has two parts (Table 3 and Supplementary Fig. S3). The first proves, for any model that satisfies stated hypotheses, the mathematical results on which each step of the method relies (Supplementary Table S4, steps 1–22). The second establishes those hypotheses for concrete benchmark models. For this, each PEtab problem was translated automatically into a Lean file (`lean_translation/sbml_to_lean.py` of the reproducibility package): the rate laws and stoichiometry of the SBML model and the nominal parameter values and initial conditions of the PEtab tables are written as Lean expressions in which every number is an exact rational. Verified checkers, Boolean functions proved sound in Lean, then test that every rate law is defined and smooth on a domain containing the trajectory, that concentrations cannot become negative (quasi-positivity) and that the growth of the vector field is bounded so that the solution exists on the simulated horizon (linear growth, or a Riccati bound on a finite horizon for Crauste). The checkers are evaluated by the Lean kernel itself (`decide +kernel`), so their results are part of the proof, and each model ends with a theorem stating that it is smooth, that its solution exists and remains non-negative on the horizon, and that the solution is differentiable with respect to the parameters, which are the hypotheses of steps 14–22. This was achieved for 28 benchmark models (27 of the 34 problems analysed, and Froehlich); the reasons why the other seven problems were not verified are given in Supplementary Note *Verified benchmark models*. Because the theorems are only as relevant as the translation is faithful, the translated vector fields, parameter values and initial values were compared independently with the SBML models evaluated by libRoadRunner at the nominal state and at five random states; they agree to rounding error (largest relative difference 10^{-12}), and changing a single number of a Lean file by 1% is detected. All files compile from scratch without `sorry`; `#print axioms` reports only the standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`) for all 309 theorems and lemmas and the 28 model theorems, and every module is accepted by the independent checker `lean4checker`. The file `NonVacuity.lean` gives, for each central theorem, a concrete instance in which every hypothesis is proved (including a nonlinear residual for the identifiability theorems), so that no central statement is vacuous, and `DEFINITIONS_AUDIT.md` compares each Lean definition with the paper and the code. The numerical analyses

of this paper (J, R_{var}, κ, VIF, $\cos\Delta$, e_{fit} and the validation) were carried out in Python in double precision; a formal conclusion applies to a numerical result when the model's hypotheses are verified in Lean and the computed quantities satisfy the inequalities that the corresponding theorem requires (Supplementary Table S5).

Further verified statements. Besides Theorems 1 and 2, the first part proves that, to first order, fixing the non-selected parameters changes the outputs by at most $\sqrt{(1 - R_{\mathrm{var}}(S))\, E}\, \|\Delta\theta\|$, where $E = \sum_j E_j$ (by the Cauchy–Schwarz inequality applied to the discarded columns of J), which is the error bound behind level 3; that re-estimating more parameters never increases the fitting error and that $e_{\mathrm{fit}} \leq e_{\mathrm{rel}}$, so that the admissibility criterion is monotone when the re-estimation reaches its optimum (with a local optimiser this holds only approximately, which is why monotonicity was also checked empirically); that the greedy selection keeps κ and VIF below their thresholds and that discarding a rejected candidate permanently is correct; that bounds on κ and VIF of the unit-length columns imply a unique and stable solution of the linearised least-squares problem for the retained parameters (which is not a proof of identifiability of the nonlinear model); that error bounds on the states transfer to the measured outputs, including logarithmic transformations; and that the first-order e_{fit} differs from the exact one by a term of second order in the absolute error (which does not by itself bound the relative error used in the decision). Explicit versions of the error bound and of the cosine for finite perturbations are also proved; they bound the second-order remainder through a Lipschitz constant L_c computed from bounds on the first and second derivatives of the vector field. Evaluated for the verified models, L_c ranged from about 10^2 to 10^{24}, too large to certify any scenario at $\pm 5\%$, so these finite certificates are of theoretical value only. Theorem `practically_identifiable_grad` (step 32) is stated in the Methods section *Identifiability of the selected parameters*; its hypothesis on the nonlinear remainder is required for all parameter values in the admissible range, whereas we evaluated it at the computed profile points. In seven problems (Armistead, Blasi, Boehm, Bruno, Rahman, Sneyd, Zheng), every selected parameter satisfied these hypotheses at all evaluated points and the model itself is verified in Lean; in the other 24 admissible problems identifiability rests on the computed profiles alone. Supplementary Table S42 lists, for every claim of the Results that refers to the formal development, the Lean theorem, its hypotheses and the numerical quantities that instantiate them.

Use of large language models

An AI assistant (Claude, Anthropic) was used to help write and debug analysis code and Lean proofs and to edit the manuscript text. All code, proofs and results were checked by the authors, who take full responsibility for the content.

Data availability. The PEtab benchmark problems analysed in this study are publicly available in the Benchmark-Models-PEtab collection (https://github.com/Benchmarking-Initiative/Benchmark-Models-PEtab, commit `fcbddf1`). All results generated in this study (selected subsets, external validation, synthetic prediction experiments, comparison with other selection criteria including the profile likelihoods, identifiability of the selected parameters, robustness,

single-parameter subsets, sensitivity to the method settings, simulator check, computation times and the checks of the Lean models), one file per benchmark problem with a data dictionary, are included in the reproducibility package archived on Zenodo (doi:10.5281/zenodo.23338175) and in the folder `reproducibility/` of https://github.com/monicammr/archivos.

Code availability. The reproducibility package (Zenodo, doi:10.5281/zenodo.23338175; folder `reproducibility/` of https://github.com/monicammr/archivos) contains the Python code of the method and of every analysis, the programs that produce every figure and table of the paper from the stored results, the translator of the SBML models into Lean, and all Lean 4 proofs (folder `lean_proofs/`, a self-contained project that compiles with a single command, `./verify.sh`, and checks that no proof uses `sorry` or additional axioms). Its manual (`README.md`) describes three independent levels of checking: (A) regenerating every figure, table and number of the paper from the stored results, without simulation, in a few minutes, which shows that the paper follows from those results; (B) compiling all proofs from scratch with `./verify.sh`, in 20 min to 2 h, which shows that every theorem is proved with no unproved assumption; and (C) re-running the analyses from the public PEtab problems and comparing the new results with the stored ones, which shows that the stored results are those the code produces (minutes per problem for the method itself, about 634 h of processor time for all analyses of the paper). Supplementary Note *Reproducibility package* describes each level in detail.

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Table 3 What the formal verification guarantees. For each step of the method, the modules of the Lean development (numbers of Supplementary Table S4), the statement proved for any model that satisfies the stated hypotheses, the condition under which it applies to a numerical result, and the results of this study to which it applies. Levels of guarantee and identifiability per problem are given in Table 1; the formal statements are listed in Supplementary Table S7, the scope of each guarantee in Supplementary Table S5, and the correspondence between each claim of this paper and the Lean theorem that supports it in Supplementary Table S42.

Step of the method (modules)	Statement proved in Lean	Applies to a result when	In this study
Ranking by energy (1, 2, 11, 13)	The first-order output error when parameters are fixed is bounded by the discarded sensitivity energy; the bound transfers to transformed outputs	$R_{\mathrm{var}}(S) \geq \tau_R$ and the model is verified	Level 3: Alkan, Perelson
Selection with κ and VIF (7–9)	The greedy subset satisfies $\kappa \leq \tau_\kappa$ and VIF $\leq \tau_{\mathrm{VIF}}$; discarding a rejected candidate permanently is correct; the linearised least-squares estimates are unique and stable	The computed κ and VIF are below the thresholds (unit-length columns for the least-squares statement)	Selection: all 31 subsets; least squares: 30 (not Rahman)
Admissibility threshold (3, 4, 20–22)	Theorem 1: rescaling an output change with cosine c leaves a relative error $\sqrt{1-c^2} = \tau_e$	Median $\cos\Delta \geq c$	Level 2 or 3: 17 subsets
Re-estimation (5, 12)	More re-estimated parameters never increase the fitting error; $e_{\mathrm{fit}} \leq e_{\mathrm{rel}}$; the first-order e_{fit} differs from the exact one by a second-order term	Always (exact e_{fit}); bound on the absolute error only (first order)	25 admissible problems exact, 6 first order and then checked nonlinearly; Beer and Lang first order; Crauste not defined
Backward elimination (6)	Theorem 2: the result is admissible and no parameter can be removed	Always	All 31 subsets
Model hypotheses (14–18, 23–31)	The solution exists on the horizon, stays non-negative and is differentiable in the parameters	Checked for each model by the Lean kernel	28 models (27 of the 34 problems)
Identifiability (32)	If t_j^* is smaller than the distance to both bounds, θ_j is practically identifiable	I_j, γ and ε computed at the profile points satisfy the inequality	Hypotheses satisfied at the evaluated profile points (not a certificate): 24 parameters; all of S in 7 problems